📊 Online Book Store Data Analysis

SQL Project Report

Name: Vishal Desai
Course: SQL Micro Course
Tools Used: PostgreSQL (PGAdmin)

-- Create Table Books--

```
> DROP TABLE IF EXISTS Books;
CREATE TABLE Books(
        Book_ID SERIAL PRIMARY KEY,
        Title VARCHAR(100),
        Author VARCHAR(100),
        Genre VARCHAR(48),
        Published_Year INT,
        Price NUMERIC(10,2),
        Stock INT
);
SELECT * FROM Books;
```

-- Create Table Customers --

```
> DROP TABLE IF EXISTS Customers;
CREATE TABLE Customers(
        Customer_ID SERIAL PRIMARY KEY,
        Name VARCHAR(100),
        Email VARCHAR(71),
        Phone VARCHAR(15),
        City VARCHAR(35),
        Country VARCHAR(53)
```

```
);

SELECT * FROM Customers;
```

-- Create Table Orders --

```
> DROP TABLE IF EXISTS Orders;

CREATE TABLE Orders(

        Order_ID SERIAL PRIMARY KEY,

        Customer_ID INT REFERENCES Customers(Customer_ID),

        Book_ID INT REFERENCES Books(Book_ID),

        Order_Date DATE,

        Quantity INT,

        Total_Amount NUMERIC(10,2)

);
```

---------- Basic Questions ----------

Q1: Retrieve all books in the 'Fiction' genre

Query:

```
SELECT *

FROM Books

WHERE genre = 'Fiction';
```

Output (Result Count): 60

Output (Sample):

	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (48)	published_year integer	price numeric (10,2)	stock integer
1	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
2	22	Multi-layered optimizing migration	Wesley Escobar	Fiction	1908	39.23	78
3	28	Expanded analyzing portal	Lisa Coffey	Fiction	1941	37.51	79

Insight:
Filters books belonging to the Fiction genre.

Q2: Count books published after 1950

Query:

```
SELECT COUNT(*)

FROM Books

WHERE published_year > 1950;
```

Output (Result Count): 292

Insight:
Shows number of modern books.

Q3: Customers from Canada

Query:

SELECT *

FROM Customers

WHERE country = 'Canada';

Output (Count): 03

Output (Sample):

	customer_id [PK] integer	name character varying (100)	email character varying (71)	phone character varying (15)	city character varying (35)	country character varying (53)
1	38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
2	415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
3	468	David Hart	stokesrebecca@gmail.c...	1234568358	Thompsonfurt	Canada

Insight:
Identifies customers from Canada.

Q4: Show orders placed in November 2023 :

Query:

SELECT *

FROM Orders

WHERE order_date BETWEEN '2023-11-01' AND '2023-11-30';

Output (Count): 25

Q5: Total stock available :

Query:

SELECT SUM(stock)AS Total_Stock

FROM Books;

Output: 25,056

	total_stock bigint
1	25056

Insight:
Total inventory available.

Q6: Most expensive book

Query:

```
SELECT *

FROM Books

ORDER BY price DESC

LIMIT 1;
```

Output:
Title = Proactive system-worthy orchestration
Price = 49.98

Insight:
Finds highest priced book.

Q7: Orders with quantity > 1

Query:

```
SELECT *

FROM Orders

WHERE quantity > 1;
```

Output (Count): 438

Insight:
Shows bulk orders.

Q8: Retrieve all orders where the total amount exceeds $20 :

Query:

```
SELECT *

FROM Orders

WHERE total_amount > 20.00;
```

Output (Count): 473

Q9: Distinct genres :

Query:

```
SELECT DISTINCT(genre)AS Available_Genres

 FROM Books;
```

Output: 07

	available_genres character varying (48)
1	Romance
2	Biography
3	Mystery
4	Fantasy
5	Fiction
6	Non-Fiction
7	Science Fiction

Insight:
Lists all unique genres.

Q10: Find the Books with lowest Stock

Query:

SELECT *

FROM Books

WHERE stock = (SELECT MIN(stock) FROM Books);

Output: 0 (5)

Q11: Total revenue:

Query:

SELECT

SUM (total_amount) AS Revenue

FROM Orders;

Output: 75,628.66

Insight:
Total revenue generated.

	revenue numeric
1	75628.66

---------- Advanced Questions ----------

Q12: Total books sold per genre:

Query:

```
SELECT b.genre,SUM(o.quantity) AS Total_Books_Sold

FROM Books b

JOIN

Orders o

ON b.book_id = o.book_id

GROUP BY b.genre;
```

	genre character varying (48)	total_books_sold bigint
1	Romance	439
2	Biography	285
3	Mystery	504
4	Fantasy	446
5	Fiction	225
6	Non-Fiction	351
7	Science Fiction	447

Insight:
Identifies top-performing genres.

Q13: Average Price of Books in 'Fantasy' Genre

Query:

```
SELECT ROUND(AVG(price),2) AS Averge_Price

FROM Books

WHERE genre = 'Fantasy';
```

Output:
Average_Price = 25.98

	averge_price numeric
1	25.98

Insight:
Calculates the average price of all Fantasy genre books, helping in pricing analysis.

Q14: Customers Who Placed At Least 2 Orders

Query:

```
SELECT c.customer_id,c.name,COUNT(o.order_id) AS Order_Count

FROM customers c

JOIN Orders o ON c.customer_id = o.customer_id

GROUP BY c.customer_id,c.name

HAVING COUNT(o.order_id) >=2

ORDER BY Order_Count DESC;
```

Output (Sample):

	customer_id [PK] integer	name character varying (100)	order_count bigint
1	364	Carrie Perez	6
2	474	Anthony Young	5
3	405	Julie Smith	4
4	485	Andrew Figueroa	4

Total rows: 139 Query complete 00:00:00.187

Insight:
Identifies repeat customers who have placed multiple orders.

Q15: Most Frequently Ordered Book:

Query:

```
SELECT o.book_id,b.title,COUNT(o.order_id) AS Order_Count

FROM books b

JOIN Orders o ON b.book_id = o.book_id

GROUP BY o.book_id,b.title

ORDER BY Order_Count DESC

Limit 1;
```

Output:
Book_ID = 88
Title = Robust tangible hardware
Order_Count = 04

	book_id integer	title character varying (100)	order_count bigint
1	88	Robust tangible hardwa...	4

Insight:
Finds the most popular book based on highest number of orders.

Q16: Top 3 Most Expensive Fantasy Books:

Query:

```
SELECT * FROM Books

WHERE genre = 'Fantasy'

ORDER BY price DESC

LIMIT 3;
```

Output (Sample):

	book_id [PK] integer	title character varying (100)	author character varying (100)	genre character varying (48)	published_year integer	price numeric (10,2)	stock integer
1	240	Stand-alone content-based hub	Lisa Ellis	Fantasy	1957	49.90	41
2	462	Innovative 3rdgeneration databa...	Allison Contreras	Fantasy	1988	49.23	62
3	238	Optimized even-keeled analyzer	Sherri Griffith	Fantasy	1975	48.97	72

Insight:
Displays the top 3 highest-priced books in the Fantasy category.

Q17: Total Quantity of Books Sold by Each Author:

Query:

```
SELECT b.author,SUM(o.quantity)AS Total_Books_Sold

FROM Books b

JOIN Orders o ON b.book_id = o.book_id

GROUP BY b.author

ORDER BY Total_Books_Sold DESC;
```

Output:

	author character varying (100)	total_books_sold bigint
1	Patrick Contreras	28
2	Melissa Taylor	27
3	Emily James	24
4	Thomas Trujillo	24
5	Ellen Doyle	23
6	Sheena Harris	23
Total rows: 314	Query complete 00:00:00.126	

Insight:
Shows which authors have the highest book sales.

Q18: Cities with Customers Spending Over $30:

Query:

```
SELECT DISTINCT(c.city),o.total_Amount

FROM Customers c

JOIN Orders o ON c.customer_id = o.customer_id

WHERE o.total_Amount > 30

ORDER BY o.total_Amount DESC;
```

Output:

	city character varying (35)	total_amount numeric (10,2)
1	Smithborough	491.50
2	Fieldsland	489.60
3	Thomaschester	489.60
4	Mendezburgh	486.70
5	Melissaside	480.30
6	Freemanland	469.70

Total rows: 443 Query complete 00:00:00.099

Insight:
Identifies cities where customers have high-value transactions.

Q19: Top 5 Customers by Total Spending

Query:

```
SELECT c.customer_id,c.name,SUM(o.total_amount) AS Total_Spent

FROM customers c

JOIN Orders o ON c.customer_id = o.customer_id

GROUP BY c.customer_id,c.name

ORDER BY Total_Spent DESC

LIMIT 5;
```

Output (Sample):

	customer_id [PK] integer	name character varying (100)	total_spent numeric
1	457	Kim Turner	1398.90
2	174	Jonathon Strickland	1080.95
3	364	Carrie Perez	1052.27
4	405	Julie Smith	991.00
5	386	Pamela Gordon	986.30

Insight:
Highlights top customers contributing most to revenue.

Q20: Remaining Stock After Orders

Query:

```
SELECT b.book_id,b.title,b.stock, COALESCE(SUM(o.quantity),0)AS Total_Sold,
        b.stock - COALESCE(Sum(o.quantity),0)AS Remaning_Stock
FROM Books b
LEFT JOIN Orders o
ON
b.book_id = o.book_id
GROUP BY b.book_id
ORDER BY b.book_id;
```

Output (Sample):

	book_id [PK] integer	title character varying (100)	stock integer	total_sold bigint	remaning_stock bigint
1	1	Configurable modular throughput	100	3	97
2	2	Persevering reciprocal knowledge user	19	0	19
3	3	Streamlined coherent initiative	27	5	22
4	4	Customizable 24hour product	8	0	8
5	5	Adaptive 5thgeneration encoding	16	8	8
6	6	Advanced encompassing implementation	2	0	2
7	7	Open-architected exuding structure	95	5	90
8	8	Persistent local encoding	84	3	81

Total rows: 500 Query complete 00:00:00.152

Insight:
Shows current inventory after fulfilling all order